

A PROJECTIVE APPROACH TO NON-DIVIDING SETS

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ABSTRACT. Let $F(N)$ be the largest size of a set $A \subseteq \{1, \dots, N\}$ in which no element divides the sum of any nonempty subset of the remaining elements. We prove $F(N) = N^{1/5+o(1)}$, matching the constructions of Erdős. The proof is the density-increment argument of Pham and Zakharov for non-averaging sets, run one dimension lower: divisibility constraints survive the projective normalization $v \mapsto v/\lambda(v)$, which drops the associated convex geometry by one dimension and moves the exponent from $1/4$ to $1/5$. The proof has been formally verified in Lean 4. The novel idea was found by GPT 5.6 Sol.

1. INTRODUCTION

For finite $A \subseteq \mathbb{Z}_{>0}$ write $\Sigma(A) = \{\sum_{a \in S} a : S \subseteq A\}$ for its set of subset sums. A finite set $A \subseteq \mathbb{Z}_{>0}$ is *non-dividing* if

$$a \nmid \sum_{s \in S} s \quad \text{for every } a \in A \text{ and nonempty } S \subseteq A \setminus \{a\},$$

and $F(N)$ denotes the maximum size of a non-dividing subset of $[N] = \{1, \dots, N\}$. Erdős, Lev, Rauzy, Sándor and Sárközy introduced the problem and proved $F(N) < 3N^{1/2} + 1$ [4]; it is Problem 131 of [1]. The previously known polynomial estimates were

$$N^{1/5} \ll F(N) \leq N^{1/4+o(1)}.$$

The lower bound is implicit in the constructions of [3, 4]. The upper bound holds because every non-dividing set is non-averaging — an averaging relation $\sum_{s \in S} s = |S|a$ is already forbidden modulo a — combined with the sharp non-averaging theorem of Pham and Zakharov [5].

Theorem 1.1. $F(N) = N^{1/5+o(1)}$.

Since the exponent $1/4$ is sharp for non-averaging sets, an improvement must use the divisibilities themselves, and the obstacle is that divisibility is destroyed by translation. Our proof is the density-increment argument of [5] run one dimension lower. A dyadic pigeonhole places the set in a shell $X \leq \lambda(v) \leq 2X$ of an integer linear functional λ ; the forbidden relations $\lambda(a) \mid \sum_{v \in S} \lambda(v)$ see only the values of λ , so they survive passage to subsets, coefficient identifications of proper homogeneous generalized arithmetic progressions (GAPs) — the only operations we ever perform — and, crucially, the normalization $\pi(v) = v/\lambda(v)$ into the hyperplane $\{\lambda = 1\}$ of dimension $d-1$. A tube-of-subset-sums argument (Theorem 3.8) forces the projected set into approximate convex position, and the convex-density increment of [5], run in the hyperplane and lifted back to the lattice by Lemma 4.2, moves the exponents from their α_d to $\sigma_d = \alpha_{d-1}$ ($d \geq 2$) and $\sigma_1 = \frac{1}{5}$.

We use [5, Corollary 5, Lemmas 1, 6, 7, 8 and 13] as stated, and their preprocessing, slab and filling steps and exponent numerology in unshifted or projective variants, indicating only the modifications; routine uniformity and bookkeeping verifications are abbreviated. Every adaptation is carried out in full in the Lean 4 formalization [8], whose only non-foundational axioms are the quoted results of [5] and the Blaschke selection and volume-continuity theorems [6].

2. SHELL SETS, STRUCTURE THEOREM, PREPROCESSING

Definition 2.1. Let $B \subseteq \mathbb{Z}^d$ be a symmetric axis-parallel box, $\lambda : \mathbb{Z}^d \rightarrow \mathbb{Z}$ a nonzero integer linear functional, and $X > 0$. A finite set $V \subseteq B$ is (λ, X) -*shell non-dividing* if

$$X \leq \lambda(v) \leq 2X \quad (v \in V), \quad \text{and} \quad \lambda(a) \nmid \sum_{v \in S} \lambda(v)$$

for every $a \in V$ and nonempty $S \subseteq V \setminus \{a\}$. For such v put $\pi(v) = v/\lambda(v)$, a point of the affine hyperplane $H_\lambda = \{y \in \mathbb{R}^d : \lambda(y) = 1\}$ of dimension $d - 1$. Define

$$\sigma_1 = \frac{1}{5}, \quad \sigma_2 = \frac{1}{4}, \quad \sigma_d = \frac{d-2}{d} \quad (d \geq 3);$$

for $d \geq 2$, σ_d is the sharp non-averaging exponent α_{d-1} of [5] in dimension $d - 1$.

A symmetric homogeneous GAP is $P = \{\sum_{i=1}^e n_i q_i : |n_i| \leq N_i\}$ with $q_i \in \mathbb{Z}^\ell$; its *dimension* is e , its *rank* is $\dim_{\mathbb{R}} \{q_1, \dots, q_e\}$, and it is *proper* if the coefficient map is injective. For $t \geq 0$, tP rescales the radii, and for proper P the coefficient identification is $\phi_P : P \rightarrow \mathbb{Z}^e$. We write $s(x) = x/(\log x)^2$ and $s(C) = s(|C|)$.

Theorem 2.2 (Conlon–Fox–Pham [2]; [5, Corollary 5]). *For every ℓ and $\beta > 1$ there are $c > 0$ and $D \in \mathbb{N}$ such that the following holds for every $C \subseteq \mathbb{Z}^\ell$ of sufficiently large size m lying in an axis-parallel box Q with $|Q| \leq m^\beta$. There are a set $\hat{C} \subseteq C$ with $|\hat{C}| \geq m - c^{-1}m/\log m$, a symmetric homogeneous GAP $P(C) \supseteq \hat{C} \cup \{0\}$ of dimension $d' \leq D$, and a set $C_0 \subseteq \hat{C}$ with $|C_0| \leq s(C)$, such that*

$$q + cs(C)P(C) \subseteq \Sigma(C_0)$$

for some q , and $cs(C)P(C)$ is proper.

Here automatically $q \in s(C)P(C)$: indeed $q \in \Sigma(C_0) \subseteq s(C)P(C)$, since $C_0 \subseteq P(C)$ and $0 \in cs(C)P(C)$. A *structural package* is a fixed choice, for each eligible C , of the constants and objects in Theorem 2.2, with $P(C)$ of minimum dimension $d(C) = \dim P(C)$; then all radii satisfy $N_i \geq 1$, and for large $|C|$ properness of $cs(C)P(C)$ implies that of $P(C)$. When dimensions range over a fixed finite set, one package per ambient dimension is fixed in advance, so all constants are uniform.

Definition 2.3. Let $V \subseteq B \subseteq \mathbb{Z}^d$ with B a symmetric coordinate box, and fix a structural package. The pair (V, B) is (δ, γ) -*irreducible* if every $D \subseteq V$ with $|D| \geq \delta|V|$ satisfies

$$d(D) = d \quad \text{and} \quad |P(D)| \geq \gamma|B|.$$

This is [5, Definition 9] with the shifts of the witnessing subsets omitted.

Lemma 2.4 (Transport of the shell functional). *Let $P \subseteq \mathbb{Z}^\ell$ be a proper homogeneous GAP with steps $q_1, \dots, q_d$ and let λ be integer-linear. Then*

$$\lambda_P(n_1, \dots, n_d) = \sum_{i=1}^d n_i \lambda(q_i)$$

satisfies $\lambda_P(\phi_P(v)) = \lambda(v)$ on P . Consequently, if $V \subseteq P$ is shell non-dividing for λ , then $\phi_P(V)$ is shell non-dividing for λ_P , with the same shell parameter.

Proof. Homogeneity gives $v = \sum n_i q_i$ whenever $\phi_P(v) = (n_1, \dots, n_d)$, and the identity follows by linearity. Every forbidden relation involves only the values of the functional, so it is preserved exactly. $\square$

Lemma 2.5 (Unshifted preprocessing). *Fix $\ell, \beta > 1, \varepsilon \in (0, 1/3)$ and $K_0 \geq 1$. Let $V \subseteq B \subseteq \mathbb{Z}^\ell$ be shell non-dividing with $|B| \leq |V|^\beta$, and let $0 < \delta \leq \frac{1}{2}$, $\gamma \leq \delta^K$ and $\gamma \geq |V|^{-1/3}$, where $K = K(\ell, \beta, \varepsilon, K_0)$ is sufficiently large. Using only subsets and coefficient identifications of proper homogeneous GAPs, one can pass to a shell non-dividing $\tilde{V} \subseteq \tilde{B} \subseteq \mathbb{Z}^{\tilde{d}}$, with a package fixed in $\mathbb{Z}^{\tilde{d}}$ relative to which $(\tilde{V}, \tilde{B})$ is (δ, γ) -irreducible, such that $\tilde{d} = O_{\ell, \beta}(1)$, the output package has polynomial-box exponent bounded in $\ell, \beta, \varepsilon, K_0$, and*

$$\begin{aligned} |\tilde{B}| &\ll |V|^{-(1-\varepsilon)(\tilde{d}-\ell)} |B|, & |\tilde{V}| &> |V|^{1-\varepsilon}, & \tilde{d} &> \ell, \\ |\tilde{B}| &\ll \rho^{K_0} |B|, & |\tilde{V}| &= \rho |V| > |V|^{1-\varepsilon}, & \tilde{d} &= \ell, \\ |\tilde{B}| &\ll |B|, & |\tilde{V}| &> |V|^{1-\varepsilon}, & \tilde{d} &< \ell, \end{aligned}$$

with implied constants depending on $\ell, \beta, \varepsilon, K_0$.

Proof. This is the unshifted analogue of [5, Lemma 10]. We run their iteration of down-, up- and shrink-moves with three modifications; the accompanying bookkeeping, abbreviated here, is carried out in full in the formalization [8]. Write $n = |V|$ and $\delta_* = \delta/2$. At an accepted stage A_j (with $A_0 = V$) put $P_j = P(A_j)$, $d_j = \dim P_j$, $A_j^\# = \phi_{P_j}(\hat{A}_j)$ and $Q_j = \phi_{P_j}(P_j)$, and stop before accepting a move of size at most $n^{1-\varepsilon/2}$ or of structural dimension exceeding a threshold $\tilde{d} > D(\ell, \beta) + 3(\beta + 2) + 1$.

Translations are omitted. A move of [5, Lemma 10] passes to $A_1 - x$ for a witness $A_1 \subseteq A_j^\#$ with $|A_1| \geq \delta |A_j^\#|$ and a shift $x \in Q_j$; here witnesses are taken with $x = 0$ only, matching Definition 2.3. This restricts the available moves and changes no estimate: the down-, up- and shrink-move bounds

$$|P_{j+1}| \leq C|P_j|, \quad |P_{j+1}| \leq Cs(A_{j+1})^{-(d_{j+1}-d_j)} |P_j|, \quad |P_{j+1}| < \gamma |P_j|$$

are supplied by [5, Lemmas 6 and 8], which apply as stated because every box arising is a symmetric coordinate box, and which also give the initial bound $|P_0| \ll n^{-(1-\varepsilon)\max(0, d_0-\ell)} |B| \leq n^{\beta+1}$.

The shell functional is transported. At each coefficient identification the functional is replaced by λ_{P_j} as in Lemma 2.4, so every accepted set remains shell non-dividing with the same shell parameter.

Packages are fixed in advance. Packages with common polynomial-box exponent $\bar{\beta} = 4(\beta + 3)$ are fixed beforehand in all ambient dimensions $r \leq \tilde{d}$, and stay applicable throughout: partial products of the move estimates give $|P_j| \leq n^{\beta+2}$, while $\gamma \geq n^{-1/3}$ and $|A_j| > n^{1-\varepsilon/2}$ give $\delta |A_j^\#| \geq \frac{1}{2} n^{2/3-\varepsilon/2}$, whose $\bar{\beta}$ -th power exceeds $n^{\beta+2}$ because $\varepsilon < 1/3$.

With these modifications the analysis of [5, proof of Lemma 10] applies verbatim. The total upward dimension change is at most $3(\beta + 2)$, so the dimension cutoff never triggers; a proposed iterate of size at most $n^{1-\varepsilon/2}$ would give $\delta_*^{i+1} \leq n^{-\varepsilon/2}$ and, via $\gamma \leq \delta_*^{K/2}$, force $|P_i| < 1$ once K is large, so the size cutoff never triggers either. Hence the procedure stops at an irreducible pair $(\tilde{V}, \tilde{B}) = (A_i^\#, Q_i)$ with $|\tilde{V}| > |V|^{1-\varepsilon}$. The box estimates follow from the product bound

$$|P_i| \ll \gamma^{i-|\mathcal{U}|-|\mathcal{D}|} n^{-(1-\varepsilon)M} |B|, \quad M = U + \max(0, d_0 - \ell),$$

exactly as in loc. cit.: $M \geq \tilde{d} - \ell$ gives the first case, and discarding factors at most one gives the third; for $\tilde{d} = \ell$, the bounds $|\mathcal{U}| + |\mathcal{D}| \leq 2M$ and $n^{-(1-\varepsilon)M} \leq \gamma^{2M}$ (valid as $\varepsilon < 1/3, \gamma \geq n^{-1/3}$) give $|P_i| \ll \gamma^i |B| \ll_{K_0} \rho^{K_0} |B|$, using $\rho \gg \delta_*, \gamma \leq \delta_*^{K/2}$ and $K \geq 2K_0$. $\square$

3. THE TUBE ARGUMENT

Throughout this section, $B_{\mathbb{R}} = \prod_{i=1}^d [-M_i, M_i]$ with $M_i \in \mathbb{Z}_{\geq 1}$ denotes a symmetric real box, $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$, and $w_B(u) = \max_{x \in B_{\mathbb{R}}} |u(x)|$ for a linear functional u . The *irreducible*

setting means: d and $\beta > 1$ are fixed, with a structural package in ambient dimension d of polynomial-box exponent at most β , applicable to every subset of V of size at least $\delta|V|$; the pair (V, B) , $V \subseteq B \subseteq \mathbb{Z}^d$, is (δ, γ) -irreducible relative to it; and $n = |V|$ satisfies $|B| \leq n^\beta$. For finite $E \subseteq \mathbb{R}^d$ let $Z(E) = \{\sum_{v \in E} t_v v : 0 \leq t_v \leq 1\}$ be its zonotope, and let $\langle P \rangle$ denote the lattice generated by the steps of a homogeneous GAP P .

Lemma 3.1 (Index and containment). *Assume the irreducible setting, let $C \subseteq V$ satisfy $|C| \geq \delta|V|$ and $|B| \leq |C|^\beta$, and put $P = P(C)$, $s = s(C)$, $L = \langle P \rangle$. For all sufficiently large n :*

- (i) $P \subseteq \frac{4}{c}B_{\mathbb{R}} \subseteq C_{d,\beta}B_{\mathbb{R}}$;
- (ii) if $s\gamma \rightarrow \infty$, then L has rank d and $[\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}$.

Proof. Since $C_0 \subseteq B$ and $|C_0| \leq s$,

$$q + csP \subseteq \Sigma(C_0) \subseteq sB_{\mathbb{R}};$$

as $0 \in csP$, also $q \in sB_{\mathbb{R}}$, so $csP \subseteq 2sB_{\mathbb{R}}$, and for $p \in P$ and $k = \lfloor cs \rfloor \geq cs/2$, $kp \in csP$ gives $p \in (2s/k)B_{\mathbb{R}} \subseteq (4/c)B_{\mathbb{R}}$. This is the containment step of [5, Lemma 11] and proves (i).

For (ii), irreducibility gives $\dim P = d$ and $|P| \geq \gamma|B|$. Write $P = \{\sum n_i q_i : |n_i| \leq N_i\}$ with $N_i \geq 1$, and let $H = \text{span}_{\mathbb{R}}\{q_i\}$ have dimension r . Properness of csP gives $|csP| \gg_{d,\beta} s^d |P|$, while $csP \subseteq H \cap 2sB_{\mathbb{R}}$ and projection to r coordinates on which H injects give $|csP| \leq (2s)^r |B|$. Hence

$$s^{d-r} \gamma \ll_{d,\beta} 1,$$

so $r = d$ by $s\gamma \rightarrow \infty$, and the steps form a $\mathbb{Z}$ -basis of L . With M the matrix of steps,

$$\text{vol}_d(\text{conv } P) = 2^d |\det M| \prod_{i=1}^d N_i \asymp_d [\mathbb{Z}^d : L] |P|,$$

while (i) gives $\text{vol}_d(\text{conv } P) \ll_{d,\beta} \text{vol}_d(B_{\mathbb{R}}) \asymp_d |B|$; hence $[\mathbb{Z}^d : L] |P| \ll_{d,\beta} |B| \leq \gamma^{-1} |P|$. $\square$

Lemma 3.2 (Projective span). *Assume the irreducible setting, with V (λ, X) -shell non-dividing, and suppose $[s(\delta n)]^{-1} = o(\gamma)$. For all sufficiently large n ,*

$$\text{span}_{\mathbb{R}} V = \mathbb{R}^d, \quad \text{aff } \pi(V) = H_\lambda, \quad \pi(V) \subseteq \Omega := \{y \in H_\lambda : Xy \in B_{\mathbb{R}}\},$$

and Ω has nonempty relative interior in H_λ .

Proof. Since $s(V) \geq s(\delta n)$, the hypothesis gives $s(V)\gamma \rightarrow \infty$, so Lemma 3.1(ii) with $C = V$ makes the step lattice of $P(V)$ have rank d . For large n every radius of $cs(V)P(V)$ is at least one, so differences of points of $q + cs(V)P(V) \subseteq \Sigma(V_0) \subseteq \text{span}_{\mathbb{R}} V$ realize every step of $P(V)$; hence $\text{span}_{\mathbb{R}} V = \mathbb{R}^d$. If $\pi(V)$ lay in a proper affine subspace of H_λ , then $v = \lambda(v)\pi(v)$ would place V in a proper linear subspace of $\mathbb{R}^d$. Finally $X\pi(v) = \frac{X}{\lambda(v)}v \in B_{\mathbb{R}}$ by the shell condition and convexity of $B_{\mathbb{R}}$, so $\pi(V) \subseteq \Omega$; and Ω is convex and contains an affinely spanning subset of H_λ . $\square$

Lemma 3.3 (Filling). *Let $C \subseteq B \subseteq \mathbb{Z}^d$, let $P = P(C)$ have dimension d and step lattice L of rank d , and put $\bar{C} = \hat{C} \setminus C_0$ and $s = s(C)$. For all sufficiently large $|C|$,*

$$L \cap \left(Z(\bar{C}) + q + \frac{cs}{2}P \right) \subseteq \Sigma(C).$$

Proof. This inclusion is established from the rounding lemma [5, Lemma 13] in the proof of [5, Theorem 4]; we indicate the homogeneous full-rank form. The steps of P are a $\mathbb{Z}$ -basis of L , so the coefficient map extends to a linear isomorphism ψ with $\psi(P) = \prod[-N_i, N_i] \cap \mathbb{Z}^d$. Let

$y = z + t$ with $z \in Z(\overline{C})$ and $t \in q + (cs/2)P$; then $z = y - t \in L$, since $q \in sP \subseteq L$. Applying [5, Lemma 13] to $\psi(\overline{C}) \cup \{0\}$ in the coefficient box yields $\tilde{z} \in \Sigma(\overline{C})$ with

$$z - \tilde{z} \in 3\sqrt{2d|C|}P \subseteq \frac{cs}{2}P$$

for large $|C|$, because $s = |C|/(\log |C|)^2$. Then $(z - \tilde{z}) + t \in q + csP \subseteq \Sigma(C_0)$, and disjointness of $\overline{C}$ and C_0 gives $y = \tilde{z} + ((z - \tilde{z}) + t) \in \Sigma(C)$. $\square$

Lemma 3.4 (Slab exclusion). *Assume the irreducible setting, and suppose $\delta \geq n^{-1/2}$ and $[s(\delta n)]^{-1} = o(\gamma)$. There is $c_{d,\beta} > 0$ such that, for every nonzero linear functional u and every $0 < \tau \leq c_{d,\beta}\gamma$, the central slab $\{x : |u(x)| \leq \tau w_B(u)\}$ contains fewer than δn points of V .*

Proof. This is the unshifted form of the slab claim in the proof of [5, Lemma 14]. Suppose $D \subseteq V$ lies in the slab and $|D| \geq \delta n$. Since $\delta \geq n^{-1/2}$, one has $|B| \leq n^\beta \leq |D|^{2\beta}$, so the package applies to D , and irreducibility gives $d(D) = d$ and $|P(D)| \geq \gamma|B|$.

Let $K = B_{\mathbb{R}} \cap \{x : |u(x)| \leq \tau w_B(u)\}$ and let r be the dimension of $\text{span}_{\mathbb{R}}(K \cap \mathbb{Z}^d)$. The discrete John lemma [5, Lemma 7], applied to K in that span's lattice, places $K \cap \mathbb{Z}^d \supseteq D$ in a proper symmetric rank- r GAP Q with $|Q| \ll_d |K \cap \mathbb{Z}^d|$. If $r < d$, then trivially $|Q| \ll_d |B|$. If $r = d$, then moreover

$$|K \cap \mathbb{Z}^d| \ll_d \text{vol}_d(K) \leq d\tau \text{vol}_d(B_{\mathbb{R}}) \asymp_d \tau|B|,$$

by the successive-minima count [7, Ch. 3] and a section estimate: writing $u(x) = \sum a_i x_i$ and choosing j with $|a_j|M_j \geq w_B(u)/d$, the x_j -sections of K are intervals of length at most $2\tau w_B(u)/|a_j| \leq 2d\tau M_j$.

Put $s_D = s(D) \geq s(\delta n)$. Comparing the proper GAP $q + cs_D P(D) \subseteq \Sigma(D_0) \subseteq s_D Q$ gives $s_D^d |P(D)| \ll_{d,\beta} s_D^r |Q|$. For $r < d$ this yields $|P(D)| \ll s_D^{-1} |B| = o(\gamma|B|)$ by the parameter hypothesis; for $r = d$ it yields $|P(D)| \ll \tau|B| \leq Cc_{d,\beta}\gamma|B|$. Choosing $c_{d,\beta}$ small enough contradicts $|P(D)| \geq \gamma|B|$ in either case. $\square$

Lemma 3.5 (Robust weighted zonotope). *Assume the irreducible setting with $\delta \geq n^{-1/2}$, $\delta = o(\mu)$ and $[s(\delta n)]^{-1} = o(\gamma)$. Let $a \in V$, let weights $0 \leq \theta_v \leq \frac{1}{2}$ ($v \neq a$) satisfy $c_1\mu n \leq \sum_{v \neq a} \theta_v \leq c_2\mu n$, let $R \subseteq V \setminus \{a\}$ have size $o(\mu n)$, and put $C = (V \setminus \{a\}) \setminus R$ and $z_C = \sum_{v \in C} \theta_v v$. Then, for a constant $c_{d,\beta,c_1,c_2} > 0$,*

$$z_C + c_{d,\beta,c_1,c_2} \mu \gamma n B_{\mathbb{R}} \subseteq Z(C).$$

Proof. We follow the proof of [5, Lemma 14], with the slab claim replaced by Lemma 3.4. With $Z^0 = \sum_{v \in C} [-\theta_v, \theta_v]v$ it suffices to prove $\xi n B_{\mathbb{R}} \subseteq Z^0$ for $\xi = c_{d,\beta,c_1,c_2} \mu \gamma$. If $y \in \xi n B_{\mathbb{R}} \setminus Z^0$, separation gives a functional u with

$$z := u(y) > \sum_{v \in C} \theta_v |u(v)|,$$

and $H = \{x : |u(x)| < c'_d \gamma z / (\xi n)\}$ is a central slab of relative width at most $c'_d \gamma$, because $y \in \xi n B_{\mathbb{R}}$. By Lemma 3.4, $|V \cap H| < \delta n$. The mass count is then that of loc. cit.: the total weight is $\asymp \mu n$, removing R costs $o(\mu n)$, and the slab carries at most $\delta n/2 = o(\mu n)$, so $\sum_{v \in C \setminus H} \theta_v \gg \mu n$; since $|u(v)| \geq c'_d \gamma z / (\xi n)$ off H , this gives $\sum_v \theta_v |u(v)| \gg \mu n \cdot \gamma z / (\xi n) > z$ once c_{d,β,c_1,c_2} is small, a contradiction. $\square$

Definition 3.6. An indexed family $p_1, \dots, p_n$ in a Euclidean affine space is in μ -convex position if every p_j lies in a closed half-space containing at most μn indices, counted with multiplicity — the indexed variant of the δ -convex position of [5]. In a zero-dimensional space the only nonempty half-space is the whole space.

Lemma 3.7 (Projective duality). *Let V lie in a (λ, X) -shell and suppose the indexed family $\pi(V)$ is not in μ -convex position, where $\mu n \geq 4$. Then there are $a \in V$ and coefficients β_v with*

$$a = \sum_{v \neq a} \beta_v v, \quad 0 \leq \beta_v \leq \frac{4}{\mu n}, \quad \frac{1}{3} \leq \sum_{v \neq a} \beta_v \leq 3.$$

Proof. Choose an index a at which convex position fails. The separation argument opening the proof of [5, Theorem 4] yields coefficients $c_v \in [0, (\mu n)^{-1}]$ with $\sum_v c_v = 1$ and $\pi(a) = \sum_v c_v \pi(v)$: a functional separating 0 from the compact convex set of such combinations would be negative on it, while failure of convex position places more than μn indices in the half-space where it is nonnegative, and distributing the unit mass over those indices gives a contradiction. Multiplying by $\lambda(a)$ gives $a = \sum_v c_v \frac{\lambda(a)}{\lambda(v)} v$ with every ratio in $[\frac{1}{2}, 2]$; moving the $v = a$ term to the left and dividing by $1 - c_a \geq \frac{3}{4}$ gives the stated bounds. $\square$

Theorem 3.8 (Projective convexity). *Assume the irreducible setting with V (λ, X) -shell non-dividing, $0 < \delta, \gamma, \mu < 1$, and the parameter hierarchy*

$$\delta \geq n^{-1/2}, \quad \delta = o(\mu), \quad \frac{n}{\log n} = o(\mu \gamma n), \quad \gamma^{-1} = o(\mu \gamma n), \quad [s(\delta n)]^{-1} = o(\gamma).$$

Then $\pi(V)$ is in μ -convex position for all sufficiently large n .

Proof. Suppose not. Lemma 3.7 gives a and coefficients β_v ; set $T = \lfloor c_0 \mu n \rfloor$ and $\theta_v = T \beta_v$ for a small absolute constant c_0 , so that

$$Ta = \sum_{v \neq a} \theta_v v, \quad 0 \leq \theta_v \leq \frac{1}{2}, \quad \sum_{v \neq a} \theta_v \asymp \mu n.$$

Put $C = V \setminus \{a\}$, with structural objects $\widehat{C}, C_0, P, q$, and $L = \langle P \rangle$, $\overline{C} = \widehat{C} \setminus C_0$, $s = s(C)$. For large n one has $|C| \geq \delta n$ and $|B| \leq |C|^{2\beta}$, so Lemma 3.1 (applied with exponent 2β) gives $\text{rank } L = d$, $[\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}$ and $P \subseteq C_{d,\beta} B_{\mathbb{R}}$.

The discarded set $R = C \setminus \overline{C}$ has size $O(n/\log n) = o(\mu n)$, so Lemma 3.5 gives

$$\bar{z} + \xi_0 n B_{\mathbb{R}} \subseteq Z(\overline{C}), \quad \bar{z} = \sum_{v \in \overline{C}} \theta_v v, \quad \xi_0 \asymp_{d,\beta} \mu \gamma,$$

while $Ta - \bar{z} \in O(n/\log n) B_{\mathbb{R}}$ and $q \in sP$. Hence $(Ta - \bar{z}) - q \in O(n/\log n + s) B_{\mathbb{R}} = o(\xi_0 n) B_{\mathbb{R}}$ by the hierarchy, so with $\xi = \xi_0/2$, for large n , $(Ta - \bar{z}) - q + \xi n B_{\mathbb{R}} \subseteq \xi_0 n B_{\mathbb{R}}$. Consequently every $y \in L \cap (Ta + \xi n B_{\mathbb{R}})$ satisfies $y - q \in \bar{z} + \xi_0 n B_{\mathbb{R}} \subseteq Z(\overline{C})$, hence $y \in Z(\overline{C}) + q + \frac{cs}{2} P$, and Lemma 3.3 yields the tube inclusion

$$L \cap (Ta + \xi n B_{\mathbb{R}}) \subseteq \Sigma(C). \quad (\dagger)$$

Let r be the order of $a + L$ in $\mathbb{Z}^d/L$; then $r \ll_{d,\beta} \gamma^{-1}$, so $r = o(\xi n)$ and $r = o(T)$ by the hierarchy. The nearest multiple m_0 of r to T is therefore positive and satisfies $|m_0 - T| \leq r/2 < \xi n$; since $a \in B$,

$$m_0 a \in L \cap (Ta + \xi n B_{\mathbb{R}}) \subseteq \Sigma(V \setminus \{a\})$$

by $(\dagger)$. Applying λ gives $m_0 \lambda(a) = \sum_{v \in S} \lambda(v)$ for a nonempty $S \subseteq V \setminus \{a\}$, contradicting shell non-divisibility. $\square$

4. PROJECTIVE DENSITY INCREMENT

Lemma 4.1 (Weighted convex-density lemma). *Fix $r \geq 1$ and $\varepsilon > 0$. There are $\tau \in (0, 1)$ and $\mu_0 > 0$ such that the following holds for $0 < \mu < \mu_0$ and n sufficiently large in r, ε, μ . If an*

indexed family $p_1, \dots, p_n$ in a convex body $\Omega \subseteq \mathbb{R}^r$ with nonempty interior is in μ -convex position, then there are $\eta \in [\mu, \mu^\tau]$ and a compact convex $\Omega' \subseteq \Omega$ with

$$\text{vol}_r(\Omega') \leq \eta \text{vol}_r(\Omega), \quad \#\{i : p_i \in \Omega'\} \geq \eta^{\kappa_r + \varepsilon} n, \quad \kappa_r = \frac{r-1}{r+1}.$$

Proof. Take τ and μ_0 from [5, Lemma 1]. For each j choose a witnessing closed half-space H_j containing p_j and at most μn indices, and for $m \geq 1$ perturb the points successively to distinct $q_i^{(m)} \in H_i \cap B(p_i, 1/m)$, so close to p_i that $q_i^{(m)} \in H_j$ only when $p_i \in H_j$; this is possible because only finitely many half-spaces H_j omit p_i , each at positive distance, and $H_i \cap B(p_i, 1/m)$ is infinite as $r \geq 1$. The perturbed set is in μ -convex position inside $\Omega_m = \Omega + m^{-1}\overline{B_2}$, so [5, Lemma 1] gives $\eta_m \in [\mu, \mu^\tau]$ and a compact convex $K_m \subseteq \Omega_m$ with $\text{vol}_r(K_m) \leq \eta_m \text{vol}_r(\Omega_m)$ and $|\{i : q_i^{(m)} \in K_m\}| \geq \eta_m^{\kappa_r + \varepsilon} n$.

Pass to a subsequence along which $\eta_m \rightarrow \eta$, the finite index set $\{i : q_i^{(m)} \in K_m\}$ is constant, and $K_m \rightarrow K$ in the Hausdorff metric (Blaschke selection; convex-body volume is continuous under Hausdorff convergence [6]). Since Ω_m decreases to Ω , one has $K \subseteq \Omega$, every stabilized index satisfies $p_i \in K$, and $\Omega' = K$ has the required properties. $\square$

Lemma 4.2 (Cap lifting). *Let $B_{\mathbb{R}} \subseteq \mathbb{R}^d$ be a symmetric coordinate box, $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$, λ a nonzero integer linear functional, m_λ the induced Euclidean measure on H_λ , and $\Omega = \{y \in H_\lambda : Xy \in B_{\mathbb{R}}\}$. Let $\Omega' \subseteq \Omega$ be compact convex with $m_\lambda(\Omega') \leq \eta m_\lambda(\Omega)$, and put*

$$W = \{v \in B : X \leq \lambda(v) \leq 2X, \pi(v) \in \Omega'\}.$$

Then there are a symmetric coordinate box $B' \subseteq \mathbb{Z}^r$, an integer-linear functional λ' , and an injective map $\phi : W \rightarrow B'$ with $\lambda'(\phi(v)) = \lambda(v)$ on W , such that either $r < d$ and $|B'| \ll_d |B|$, or $r = d$ and $|B'| \ll_d \eta |B|$. In particular, shell non-divisibility and the shell parameter are preserved exactly.

Proof. Write $\lambda(x) = \ell \cdot x$ and, for $E \subseteq H_\lambda$ and $T > 0$, put $\text{Cone}_T(E) = \{ty : y \in E, 0 \leq t \leq T\}$. Integrating the Jacobian gives

$$\text{vol}_d(\text{Cone}_T(E)) = \frac{T^d}{d\|\ell\|_2} m_\lambda(E),$$

so the λ -dependent factor cancels from volume ratios. Let $K = B_{\mathbb{R}} \cap \text{Cone}_{2X}(\Omega')$, a convex set containing 0 and W ; since $\text{Cone}_X(\Omega) \subseteq B_{\mathbb{R}}$, the display gives $\text{vol}_d(K) \leq 2^d \eta \text{vol}_d(B_{\mathbb{R}})$.

Put $S = (K - K) \cap \mathbb{Z}^d$, $H = \text{span}_{\mathbb{R}} S$ and $r = \dim H$; then $W \subseteq S$ and $K - K \subseteq 2B_{\mathbb{R}}$. The discrete John lemma [5, Lemma 7], applied to $(K - K) \cap H$ in the lattice $H \cap \mathbb{Z}^d$, gives a proper symmetric homogeneous rank- r GAP $Q \supseteq S$ with $|Q| \ll_d |S|$. If $r < d$, then $|S| \leq |2B_{\mathbb{R}} \cap \mathbb{Z}^d| \leq 2^d |B|$. If $r = d$, the successive-minima count [7, Ch. 3] and the Rogers–Shephard inequality [6] give

$$|S| \ll_d \text{vol}_d(K - K) \leq \binom{2d}{d} \text{vol}_d(K) \ll_d \eta |B|.$$

Finally let B' be the coefficient box of Q , let $\phi = \phi_Q|_W$, and let $\lambda'(n_1, \dots, n_r) = \sum n_i \lambda(q_i)$; Lemma 2.4 gives $\lambda'(\phi(v)) = \lambda(v)$, and $|B'| = |Q|$ satisfies the stated bounds. $\square$

5. THE ITERATION AND THE PROOF OF THEOREM 1.1

Lemma 5.1 (Numerology). *Fix $D \geq 2$ and $\zeta_0 > 0$. There are $\varepsilon_0, c_* > 0$ such that, whenever $1 \leq d < e \leq D$, $\zeta_0 \leq \zeta \leq 1 - \sigma_d$ and $0 < \varepsilon < \varepsilon_0$,*

$$(\sigma_e + \zeta) \left(\frac{1}{\sigma_d + \zeta} - (1 - \varepsilon)(e - d) \right) \leq 1 - c_*$$

whenever the parenthetical factor is nonnegative.

Proof. For $2 \leq d < e$ this is [5, Observation 15] with all dimensions shifted down by one, the gap depending continuously on ζ . Only the pairs with $d = 1$ are new: for $k = e - 1$, the function

$$f_{1,k}(\zeta) = (\sigma_{1+k} + \zeta) \left(\frac{1}{\sigma_1 + \zeta} - k \right)$$

satisfies $f_{1,k}(0) = \sigma_{1+k}(5-k) \leq 1$, with equality only for $e \in \{2, 3, 4\}$, and $f'_{1,k}(\zeta) = \frac{\sigma_1 - \sigma_{1+k}}{(\sigma_1 + \zeta)^2} - k < 0$, so the gap is strictly positive for $\zeta \geq \zeta_0$. Compactness over the finitely many pairs and the range $\zeta \in [\zeta_0, 1 - \sigma_d]$ gives a uniform $c_* > 0$, which persists for ε_0 small. $\square$

Theorem 5.2. *For fixed d and $\beta > 1$, every (λ, X) -shell non-dividing set $V \subseteq B \subseteq \mathbb{Z}^d$, with B a symmetric coordinate box and $|B| \leq |V|^\beta$, satisfies $|V| \leq |B|^{\sigma_d + o(1)}$, uniformly in λ and X .*

Proof. Suppose for contradiction that for some $\zeta_0 > 0$ there are arbitrarily large examples $A_0 \subseteq B_0 \subseteq \mathbb{Z}^{d_0}$ with $|A_0| > |B_0|^{\sigma_{d_0} + \zeta_0}$; since $|A_0| \leq |B_0|$, we may assume $\sigma_{d_0} + \zeta_0 < 1$. All sets constructed below remain shell non-dividing, by Lemma 2.4.

Choice of parameters. At any stage of excess $\zeta \geq \zeta_0$ one has $|B| < |A|^{\beta_*}$ with $\beta_* = (\sigma_1 + \zeta_0)^{-1}$, and the stage dimension is at most $D_0 = \max\{r : \sigma_r \leq 1 - \zeta_0\}$. Choose D above D_0 and above every output-dimension bound of Lemma 2.5 for dimensions $\leq D_0$ with exponent β_* ; let ε_0, c_* be as in Lemma 5.1 for (D, ζ_0) , and put $\Delta_D = \min_{2 \leq r \leq D} (\sigma_r - \sigma_{r-1})$. Choose

$$\varepsilon < \min\left\{\varepsilon_0, \frac{\zeta_0}{4}, \frac{c_*}{4}, \frac{\Delta_D}{8}, \frac{1}{3}\right\} \quad \text{with} \quad \kappa_{r-1} + \varepsilon < \sigma_r + \zeta_0 \quad (2 \leq r \leq D),$$

choose K_0 with $K_0^{-1} < \sigma_1 + \zeta_0$ and $K_0(\sigma_1 + \zeta_0) > 2$, let K be admissible in Lemma 2.5 for all $\ell \leq D_0$, let $\beta^\dagger$ bound the output package exponents and $2\beta_*/(1 - \varepsilon)$, and fix a small $\kappa \in (0, 1)$. With $N_r(t)$ a largeness threshold for Lemma 4.1 in dimension r at parameter $\mu = t$, choose a nondecreasing $\omega(n) \rightarrow \infty$ with

$$\omega(n)^{2K+2} = o(\log n), \quad N_r(\omega(n)^{-\kappa}) \leq n^{1-\varepsilon} \quad (1 \leq r \leq D-1).$$

At a stage of size n use $\delta = \omega(n)^{-1}$, $\gamma = \delta^K$, $\mu = \delta^\kappa$; then, uniformly,

$$\delta \geq n^{-1/2}, \quad \gamma \geq n^{-1/3}, \quad \delta = o(\mu), \quad \frac{1}{\log n} = o(\mu\gamma), \quad \gamma^{-1} = o(\mu\gamma n), \quad [s(\delta n)]^{-1} = o(\gamma),$$

routine verifications carried out in the formalization [8].

The one-step increment. At a stage $A \subseteq B \subseteq \mathbb{Z}^d$ with $n = |A|$, let $p = \frac{\log n}{\log |B|} = \sigma_d + \zeta$, so $\zeta_0 \leq \zeta \leq 1 - \sigma_d$ and $d \leq D_0$. Apply Lemma 2.5: the output $\tilde{A} \subseteq \tilde{B} \subseteq \mathbb{Z}^e$ has $\tilde{n} = |\tilde{A}| > n^{1-\varepsilon}$, the displayed parameter conditions persist with $\tilde{n}$ in place of n , $|\tilde{B}| \leq \tilde{n}^{\beta^\dagger}$, and $\tilde{n} \geq N_r(\mu)$ for all $r < D$.

The dimension-changing cases are handled exactly as Cases 1–2 of the proof of [5, Theorem 2], with [5, Observation 15] replaced by Lemma 5.1. If $e > d$, put $a = \frac{1}{p} - (1 - \varepsilon)(e - d)$; the box estimate gives $|\tilde{B}| \ll n^a$, impossible for large n when $a \leq 0$, and otherwise $(\sigma_e + \zeta)a \leq 1 - c_*$ yields

$$|\tilde{A}| > n^{1-\varepsilon} > |\tilde{B}|^{\sigma_e + \zeta + c_{\text{up}}}$$

for a fixed $c_{\text{up}} > 0$; any noncontradictory output has $\sigma_e + \zeta + c_{\text{up}} < 1$, hence $e \leq D_0$. If $e < d$, then $|\tilde{B}| \ll |B|$ and $\sigma_d - \sigma_e \geq \Delta_D$ give

$$|\tilde{A}| > |\tilde{B}|^{\sigma_e + \zeta + c_{\text{down}}}$$

for a fixed $c_{\text{down}} > 0$.

Suppose $e = d$, and write $|\tilde{A}| = \rho n$ and $|\tilde{B}| \ll \rho^{K_0} |B|$. If $d = 1$, Theorem 3.8 is already contradictory: the hyperplane $H_{\tilde{\lambda}}$ is zero-dimensional, its only nonempty half-space contains every index, and $\mu < 1$. Let $d \geq 2$. By Lemma 3.2, the family $\pi(\tilde{A})$ affinely spans $H_{\tilde{\lambda}}$ and lies in the convex body Ω , which has nonempty relative interior. Theorem 3.8, Lemma 4.1 applied in $H_{\tilde{\lambda}} \cong \mathbb{R}^{d-1}$, and Lemma 4.2 then produce a shell non-dividing $A' \subseteq B'$ with

$$|A'| \geq \rho \eta^{a_d} n, \quad a_d = \kappa_{d-1} + \varepsilon, \quad \eta \in [\mu, \mu^{\hat{\tau}_d}], \quad \hat{\tau}_d = \tau_{d-1} > 0.$$

If cap lifting lowers the dimension to $f < d$, then $|B'| \ll |\tilde{B}| \ll |B|$ and $|A'| \geq n^{1-2\varepsilon}$ (as $\eta^{a_d} = n^{-o(1)}$), and the computation of the case $e < d$ gives $|A'| > |B'|^{\sigma_f + \zeta + c_{\text{down}}}$. If it preserves the dimension, then $|B'| \ll \eta \rho^{K_0} |B|$ and

$$\frac{|A'|}{|B'|^p} \gg \eta^{a_d - p} \rho^{1 - K_0 p}.$$

Uniformly for $2 \leq d \leq D$ one has $p - a_d \geq \nu_0 > 0$ and $K_0 p \geq 2$, so with $\tau_* = \min_{2 \leq r \leq D} \hat{\tau}_r$ the right-hand side is at least $C_D^{-1} \mu^{-\tau_* \nu_0} \rightarrow \infty$: the excess strictly increases, and moreover $|B'|/|B| \ll_D \eta \rho^{K_0} \leq C_D \mu^{\tau_*} = o(1)$.

Termination. Upward steps increase the excess by c_{up} while the excess stays below 1, so there are $O_{D, \zeta_0}(1)$ of them; downward steps decrease the dimension, so there are at most $M = M(D, \zeta_0)$ dimension-changing steps in all, each retaining at least $|A_j|^{1-2\varepsilon}$ points. For a maximal block of same-dimensional steps in dimension r , put $c_r = \max\{\kappa_{r-1} + \varepsilon, K_0^{-1}\} < \sigma_r + \zeta_0$ and, at step j , $x_j = \log(1/\eta_j)$, $y_j = \log(1/\rho_j)$. The point and box estimates give

$$\log \frac{|A_j|}{|A_{j+1}|} \leq (\kappa_{r-1} + \varepsilon)x_j + y_j \leq c_r(x_j + K_0 y_j), \quad \log \frac{|B_j|}{|B_{j+1}|} \geq x_j + K_0 y_j - O_D(1),$$

where the $O_D(1)$ term is $o(x_j + K_0 y_j)$, because $x_j \geq \tau_* \kappa \log \omega(|A_j|) \rightarrow \infty$ as long as all cardinalities exceed a fixed power of $n_0 = |A_0|$. Summing over any initial segment of the block,

$$\log \frac{|A_{\text{start}}|}{|A_{\text{end}}|} \leq (c_r + o(1)) \log |B_{\text{start}}|, \quad \text{whence} \quad |A_{\text{end}}| \geq |A_{\text{start}}|^{1 - c_r/(\sigma_r + \zeta_0) - o(1)} \geq |A_{\text{start}}|^\theta$$

for a fixed $\theta > 0$. With at most M dimension changes and $M + 1$ blocks, every iterate satisfies $|A_j| \geq n_0^q$ for a fixed $q > 0$; running the construction with the provisional stopping rule $|A_j| < n_0^{q/2}$ makes the $o(1)$ terms uniform and shows the rule never triggers. The iteration therefore never stops; but after the last dimension change every step multiplies the box cardinality by a factor $o(1) < \frac{1}{2}$, while the sets keep at least $n_0^q > 1$ elements. This is the required contradiction. $\square$

Proof of Theorem 1.1. The lower bound $F(N) \gg N^{1/5}$ is implicit in the constructions of [3, 4] (see also [1]): translating a Bosznay-type non-averaging set of size m and diameter D by any $x > mD - \min B$ turns divisibility relations into averaging relations, and optimizing the parameters places such a set in $[2m^5]$.

For the upper bound, fix $0 < \varepsilon < 1$ and suppose, for arbitrarily large N , that some non-dividing $A \subseteq [N]$ has $|A| > N^{1/5+2\varepsilon}$. Some dyadic interval $[X, 2X)$ contains $A_0 \subseteq A$ with $|A_0| \geq |A|/(1 + \log_2 N) > N^{1/5+\varepsilon}$. Take $d = 1$, $B = [-2X, 2X] \cap \mathbb{Z}$ and $\lambda(t) = t$; then A_0 is shell non-dividing, $|B| \leq 4N + 1$, and for large N ,

$$|A_0| > |B|^{1/5+\varepsilon/2}, \quad |B| < |A_0|^\beta, \quad \beta = (1/5 + \varepsilon/2)^{-1} > 1.$$

Theorem 5.2 gives $|A_0| \leq |B|^{1/5+o(1)}$, a contradiction for large N . $\square$

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